Earnshaw and the ring saddle

Specialised local-field tools; empty region, static electrostatics, no active feedback.

The precise no-trapping statement

$$\nabla^2 V = 0 \quad \text{in a source-free region}$$

A nonconstant harmonic potential has no local maximum or minimum in an open source-free region. A free positive charge would need a minimum of V; a free negative charge would need a maximum. Neither is available for strict static trapping.

WHY / The maximum principle is stronger than a second-derivative test and also handles points with vanishing Hessian.

WATCH / Not every equilibrium is a nondegenerate quadratic saddle: a constant-field-free region can be neutral, and higher-order stationary points exist.

Hessian trace and signs

$$V \simeq V_0 + \tfrac{1}{2}\sum_i \kappa_i x_i^2$$

$$\kappa_x + \kappa_y + \kappa_z = 0$$

At a stationary point, diagonalise the symmetric Hessian. If a nonzero eigenvalue bends up, at least one other must compensate by bending down. The energy curvatures are q times these eigenvalues.

WHY / The Laplacian is the trace in any orthonormal axes; its zero value constrains all directional curvatures together.

WATCH / All second derivatives need not be strictly positive at a general strict minimum. Use the maximum principle when the quadratic test is inconclusive.

Ring centre expansion

$$V \simeq \frac{kQ}{R}\left(1 - \frac{z^2}{2R^2} + \frac{s^2}{4R^2}\right)$$

$$E_s \simeq -kQs/(2R^3)$$

For a positive ring in the xy-plane, axial curvature is -kQ/R³. Rotational symmetry gives equal x and y curvatures, each +kQ/(2R³). Thus a negative charge is axially stable but transversely unstable.

WHY / Two equal transverse curvatures must cancel the single axial curvature in the trace.

WATCH / $s^2 = x^2 + y^2$ is distance squared from the axis, not a spherical radius squared. This is a near-centre expansion.

Time scales and legitimate exceptions

$$\omega_z^2 = k|qQ|/(mR^3)$$

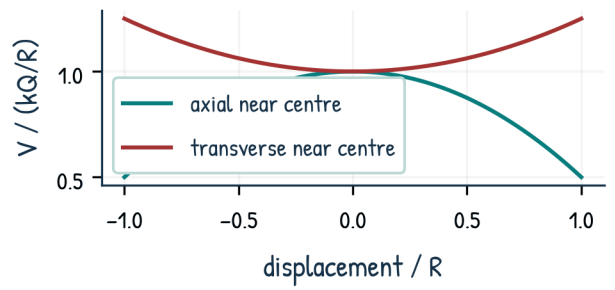
$$\tau_\perp = \sqrt{2}/\omega_z$$

For opposite-sign charge, the transverse instability grows exponentially at small displacement with this time scale. Mechanical guides, time-dependent trapping fields and feedback change the problem; magnetic/material forces are outside the theorem.

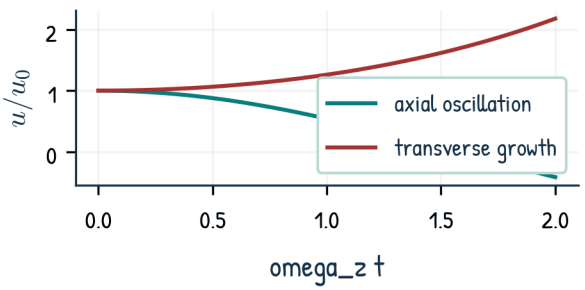
WHY / A restoring axial direction coexists with unstable transverse directions rather than violating the no-trapping condition.

WATCH / The theorem concerns static electrostatic forces on a freely moving test charge in a source-free region, not every possible levitation mechanism.

Ring centre expansion: axial downward curvature, half-size transverse upward curvature



u0 is initial displacement: stable axial vs unstable transverse motion



REMEMBER / Axial stability alone is not trapping. Harmonic potentials cannot have a strict interior extremum.

Images I: a grounded conducting plane

Charge q at (0,0,h), h>0; actual physical domain z>0; plane is at V=0.

Construction plus uniqueness

$$V = kq\left(\frac{1}{|\mathbf{r} - h\hat{\mathbf{z}}|} - \frac{1}{|\mathbf{r} + h\hat{\mathbf{z}}|}\right)$$

Replace the plane, only for the upper-half-space calculation, by an image -q at z=-h. Distances match on z=0, making V zero there. The source singularity and far boundary condition also match.

WHY / Uniqueness says this candidate is the physical exterior potential once sources and boundary values agree.

WATCH / Do not continue the image field into the conductor. The true electrostatic field in the metal is zero.

Force and the moving-image factor

$$F_z = -kq^2/(4h^2)$$

$$U(h) = -kq^2/(4h)$$

The physical charge feels attraction to its image a distance 2h away. Integrating the actual force gives the energy relative to infinity. The energy is half the naive two-real-charge interaction because induced charge readjusts.

WHY / Differentiate U with respect to h to recover Fz. This is a safe check on the one-half factor.

WATCH / Using -kq²/(2h) as physical energy would double the force when h changes and the image moves with it.

Surface charge distribution

$$\sigma(s) = -\frac{qh}{2\pi(s^2 + h^2)^{3/2}}$$

$$\int_0^\infty \sigma(s) 2\pi s \, ds = -q$$

Normal differentiation of the exterior potential gives the induced density. It is concentrated under the charge but extends across the whole infinite plane. Its total is exactly -q, supplied by ground.

WHY / The integral is finite despite the infinite plane; density falls as inverse cube at large lateral distance.

WATCH / A finite isolated plate generally induces a different total and far field. Grounded and isolated boundary conditions are not interchangeable.

Energy and scale estimate

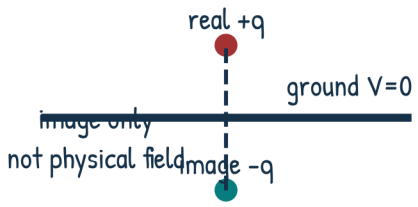
$$|U_e| = ke^2/(4h) \simeq \frac{0.360\ \text{eV nm}}{h}$$

For an electron at h=1 nm, the ideal image attraction energy is about 0.36 eV and force about 5.77 times 10^(-11) N. The continuum metal model is a macroscopic boundary approximation.

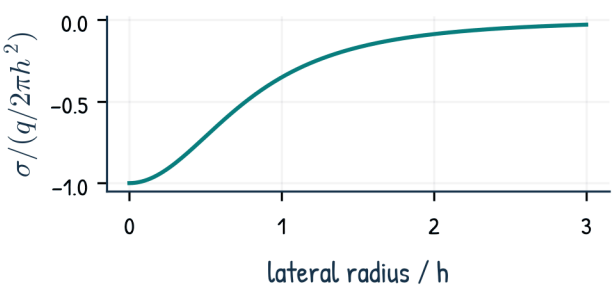
WHY / A positive or negative external charge is attracted because the induced response reverses sign with it, leaving force proportional to q².

WATCH / At atomic separations, electronic structure and quantum effects prevent extrapolating an ideal image singularity as an exact material prediction.

Grounded plane: mirror -q makes the plane an equipotential



Grounded plane: induced charge density peaks beneath the real source



REMEMBER / Images are mathematical sources outside the physical solution region, not real charges in the metal.

Images II: spherical conductor boundaries

External q lies at z=d>R outside a spherical conductor of radius R.

Grounded sphere image

$$q' = -qR/d, \quad b = R^2/d$$

$$V = kq/|\mathbf{r} - d\hat{\mathbf{z}}| + kq'/|\mathbf{r} - b\hat{\mathbf{z}}|$$

An internal image at z=b has the right distance ratio to make V zero on every point of r=R. It lies outside the physical exterior region. At infinity the potential also vanishes.

WHY / The sphere boundary sets an algebraic distance identity, and uniqueness selects the corresponding exterior solution.

WATCH / Only the exterior field is represented. Inside the real metal E is zero, not the image-pair field.

Grounded attraction and energy

$$F_z = -\frac{kq^2Rd}{(d^2 - R^2)^2}$$

$$U = -\frac{kq^2R}{2(d^2 - R^2)}$$

The source-image separation is d-R²/d. Coulomb force on q gives the stated attractive force. Half q times induced potential gives the physical energy with the grounded sphere allowed to readjust.

WHY / At d=R+h with h much smaller than R, U tends to -kq²/(4h), recovering the local plane result.

WATCH / Do not differentiate a two-real-charge energy while incorrectly holding the image charge or position fixed.

Isolated neutral sphere

$$q_0 = +qR/d \quad \text{at the centre}$$

$$Q_{\text{induced}} = q' + q_0 = 0$$

Add a central image to make total sphere charge zero while preserving a constant boundary potential. The central term makes that potential nonzero. The actual density polarises with zero net charge.

WHY / A central image shifts V equally over a spherical boundary without changing its equipotential character.

WATCH / The force is the sum of both image forces. The grounded-sphere result alone is wrong for an isolated neutral sphere.

Isolated sphere of specified charge

$$q_0 = Q_c + qR/d$$

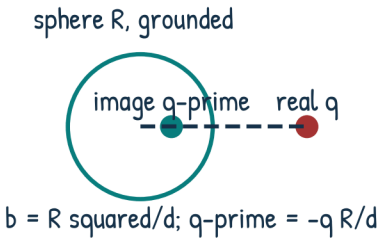
$$V_c = kq_0/R$$

The off-centre image q-prime still enforces the spatial cancellation on the sphere, and the central image sets its specified net charge Qc. Evaluate force at the actual q using both images.

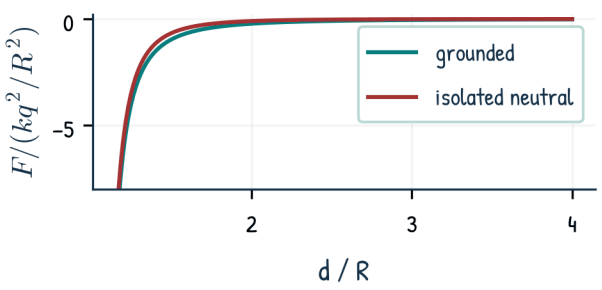
WHY / Boundary equipotential and net conductor charge are independent constraints; one image condition need not impose both.

WATCH / Do not label q-prime as the entire physical induced charge on an isolated sphere. Total charge equals the sum of all required interior images.

Grounded sphere: internal mathematical image, exterior physical solution



Both spheres polarise; grounding and neutrality give different forces



REMEMBER / Check the boundary potential first, then impose grounded versus isolated charge conditions.

Pressure, hemispheres and charged bubbles

Vacuum exterior; conducting spherical interfaces; gamma is surface tension per liquid-air surface.

Pressure from two independent routes

$$P = \sigma^2/(2\varepsilon_0)$$

$$P 4\pi R^2 = -(\partial U/\partial R)_Q$$

The patch-force route uses the field of the rest of the conductor, sigma/(2 epsilon0). The energy route differentiates U=kQ²/(2R) at fixed Q. Both give outward pressure regardless of charge sign.

WHY / Agreement checks both the field averaging and the mechanical work sign.

WATCH / At fixed potential a source exchanges energy and charge. Differentiating stored energy alone with the fixed-Q rule is inappropriate.

Separating the hemispheres

$$F_{\text{hemi}} = P\pi R^2$$

$$F_{\text{hemi}} = kQ^2/(8R^2)$$

For a uniformly charged complete spherical conducting shell, integrate the axial component P cos theta over one hemisphere. The projected area is a disc, not the hemisphere's full curved area.

WHY / Transverse force components cancel by rotational symmetry; only the cut-normal component remains.

WATCH / This is the force across the cut of the original complete shell. Widely separated hemispheres redistribute charge and do not retain this same pressure pattern.

Soap bubble: pressure ledger

$$p_{\text{in}} - p_{\text{out}} = 4\gamma/R - Q^2/(32\pi^2\varepsilon_0 R^4)$$

Surface tension from the two interfaces pulls inward; electrostatic stress pulls outward. The gas pressure difference supplies whatever residual balance is needed for a spherical static shape. For fixed gas content include its equation of state.

WHY / A charged bubble can have less gas overpressure than an uncharged bubble at the same radius.

WATCH / Zero overpressure does not imply that a punctured bubble is dynamically stable or survives. Film drainage and shape modes are separate issues.

Zero overpressure: a scale, not a verdict

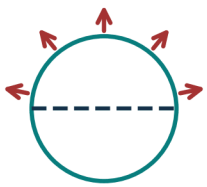
$$Q_0 = 8\pi\sqrt{2\varepsilon_0\gamma R^3}$$

For R=2 cm and gamma=0.025 N/m, the spherical zero-overpressure charge is about 47.3 nC and potential about 21.3 kV. Compare electrostatic pressure and capillary pressure with consistent units.

WHY / Both scale laws arise from equating a R^(-4) electrical pressure at fixed Q to a R^(-1) capillary pressure.

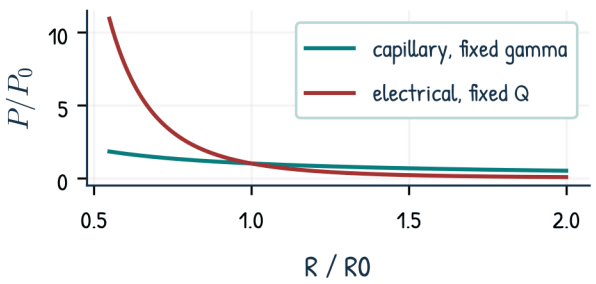
WATCH / This condition is not the liquid-drop Rayleigh instability threshold; the soap-bubble film has two surfaces and different material constraints.

Radial electrical stress: axial components over a hemisphere add



hemisphere force = pressure x projected disc area

Pressure scales differ: capillary R^-1 versus electrical R^-4



REMEMBER / Pressure is local; the net force uses projected area. Soap bubbles have TWO interfaces.

Drops: shape instability and coalescence

Conducting liquid drop in a surrounding medium approximated as vacuum; fixed volume and charge.

Rayleigh limit and its assumptions

$$Q_R = 8\pi\sqrt{\varepsilon_0 \gamma R^3}$$

$$X = Q^2/(64\pi^2\varepsilon_0\gamma R^3)$$

For an ideal isolated incompressible conducting liquid drop, the first quadrupole shape mode loses stability at fissility X=1. For water with gamma=0.072 N/m and R=1 mm, QR is about 0.635 nC.

WHY / The coefficient is a result of a volume-preserving shape perturbation. A pressure-scale comparison is suggestive but not a stability derivation.

WATCH / Gas pressure can balance a spherical drop while shape modes are unstable. Do not replace a mode analysis by an unqualified pressure equality.

Size scales at the limit

$$V_R = kQ_R/R \propto R^{1/2}$$

$$E_R = kQ_R/R^2 \propto R^{-1/2}$$

Smaller drops require less total charge to reach the Rayleigh limit but have stronger local limiting field. At R=10 micrometres, the water-drop QR is about 0.635 pC and V about 571 V.

WHY / Surface tension and electrostatic energy have different size powers, creating a nontrivial radius dependence.

WATCH / A universal 3 MV/m air cutoff is not reliable for microdroplets. Real breakdown, evaporation and ion emission depend on the environment.

Two equal drops merge: endpoint energy

$$R_f = 2^{1/3} r, \quad Q_f = 2q$$

$$\Delta U = 4\pi\gamma r^2(2^{2/3} - 2) + \frac{kq^2}{r}(2^{2/3} - 1)$$

Compare two initially FAR-SEPARATED drops with one final sphere, conserving volume and charge. Surface energy falls but electrostatic self-energy rises. Neglect initial mutual interaction only at sufficiently large separation.

WHY / The new sphere has less total area, but holding doubled charge on its larger radius still costs electrical self-energy.

WATCH / Drops initially touching have mutual interaction and deformed shapes; this far-separated endpoint formula does not establish the barrier or dynamics at contact.

Energetic threshold is not a trajectory

$$\frac{q_{\text{threshold}}}{Q_R(r)} = \frac{1}{2}\sqrt{\frac{2 - 2^{2/3}}{2^{2/3} - 1}} \simeq 0.419$$

Below this threshold the final single sphere has lower energy than two far-separated equal drops in the stated model. At r=10 micrometres for water the threshold is about 0.266 pC.

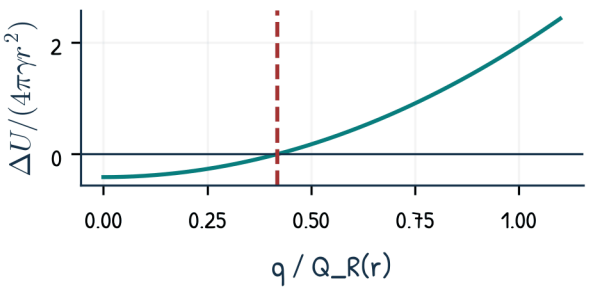
WHY / Dimensionless ratios cancel radius and material constants when both energy terms use consistent scaling.

WATCH / Lower final energy does not guarantee coalescence, nor does a higher endpoint energy prove touching drops must bounce. Barriers and dissipation matter.

Drop merging conserves volume and total charge, not surface area



Two far-separated drops to one sphere: endpoint-energy threshold about 0.419



REMEMBER / Radial pressure balance alone is not a proof of shape stability or a coalescence prediction.

Multipoles and induced neutral bodies

Localised fixed charge distributions; observation distance large relative to source size.

First terms of a far-field expansion

$$V \simeq kQ/r + k\mathbf{p}\cdot\hat{\mathbf{r}}/r^2$$

$$Q = \int dq, \quad \mathbf{p} = \int \mathbf{r}\prime\, dq$$

Expand inverse source-receiver distance in source size divided by r. Q gives the monopole. If Q=0, the dipole often leads. If both vanish, quadrupole or higher terms control the field.

WHY / Each cancellation removes the lowest distance power and exposes finer structure of the distribution.

WATCH / The dipole moment is origin-independent for a neutral distribution, but changes under origin shifts when Q is nonzero.

A simple quadrupole along its axis

$$(+q,\ -2q,\ +q) \text{ at } (-a, 0, a)$$

$$V_{\text{axis}} \simeq 2kqa^2/r^3$$

The net charge and dipole moment both vanish. On the positive axis r much greater than a, expand the three point potentials through second order. The resulting field component falls as 6kqa²/r⁴.

WHY / Symmetry cancels odd powers, while the zero total charge cancels the leading monopole.

WATCH / The axial coefficient does not describe every angular direction. Quadrupole potential has an angular pattern, not spherical symmetry.

Induced dipole: why energy has a half

$$\mathbf{p} = \alpha\mathbf{E}$$

$$U = -\tfrac{1}{2}\alpha E^2, \quad \mathbf{F} = \tfrac{1}{2}\alpha\nabla E^2$$

Creating a dipole costs internal polarisation energy p²/(2alpha). Add its interaction -p dot E and minimise over p to obtain p=alpha E and the effective -alpha E²/2.

WHY / The factor half arises because the induced moment grows as the field is established rather than already existing unchanged.

WATCH / Use the permanent-dipole energy -p dot E only when p is independently fixed. Do not insert p=alpha E after differentiating an inconsistent energy.

Small dielectric sphere and lifting estimate

$$\alpha = 4\pi\varepsilon_0 a^3 \frac{K-1}{K+2}$$

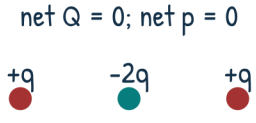
$$|F| = 2\alpha k^2 Q^2/d^5$$

The first result is for a small linear isotropic dielectric sphere in vacuum, field approximately uniform over it. In a point-source field use the local E and its gradient; attraction follows for positive alpha.

WHY / The steep d^(-5) force explains why neutral scraps respond strongly only near a charged object.

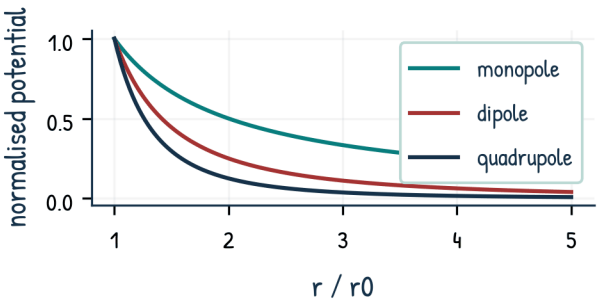
WATCH / Paper is not a perfect isotropic sphere. Geometry, humidity, contact forces and nonuniform material response make numerical lift predictions approximate.

Symmetric linear quadrupole: monopole and dipole terms cancel



first surviving far potential is quadrupolar

Successive multipoles fall faster; coefficients and angular factors differ



REMEMBER / Net neutrality removes the monopole, not all fields. An induced dipole has an assembly cost.

Off-axis rings, cube scaling and numerical checks

Use full geometry when axis-only formulas or Gauss do not determine the desired field.

Ring potential away from its axis

$$V(s,z)=\frac{kQ}{2\pi}\int_0^{2\pi}\frac{d\phi}{\sqrt{R^2+s^2+z^2-2Rs\cos\phi}}$$

An off-axis point has a source distance depending on phi, so equal-distance simplification is lost. Integrate the full scalar kernel. At s=0 recover kQ/sqrt(R²+z²); far away recover kQ/r.

WHY / Rotational symmetry reduces the spatial variables to cylindrical s,z but does not remove the angular source integral.

WATCH / Near the ideal wire, kernels become sharply peaked and eventually singular. A coarse angular grid can appear converged while being inaccurate.

Elliptic representation and checks

$$V=\frac{2kQ}{\pi D}K(m)$$

$$D=\sqrt{(R+s)^2+z^2},\quad m=4Rs/D^2$$

Here K(m) is the complete elliptic integral with PARAMETER m. Some conventions instead label the function by modulus sqrt(m). At s=0, K(0)=pi/2 gives the known axial potential.

WHY / The elliptic function is a compact name for the same source integral, not additional electrostatic physics.

WATCH / Distinguish parameter and modulus conventions. At the ideal wire m approaches one and potential diverges logarithmically.

Uniform cube: a no-integral ratio

$$V_{\mathrm{centre}}(L)=2V_{\mathrm{corner}}(L)$$

$$V\propto k\rho L^2$$

Tile a cube of side 2L with eight cubes of side L. At the large centre each small cube contributes its corner potential. Scaling fixed density gives Vcorner(2L)=4Vcorner(L), hence the ratio two.

WHY / The exact ratio uses superposition and geometric similarity even though Gauss cannot directly give the cube field.

WATCH / Density is held fixed in the scaling argument. Holding total charge fixed would change the size power.

Numerical integrals: independent checks

$$V_{\mathrm{corner}}\simeq 1.19004\,k\rho L^2$$

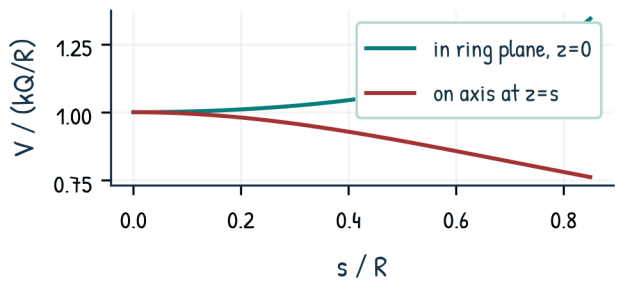
$$V_{\mathrm{centre}}\simeq 2.38008\,k\rho L^2$$

For a unit cube, integrate 1/distance with a converged quadrature or a singularity-aware transformation. The centre field is exactly zero by inversion symmetry even while its potential is nonzero.

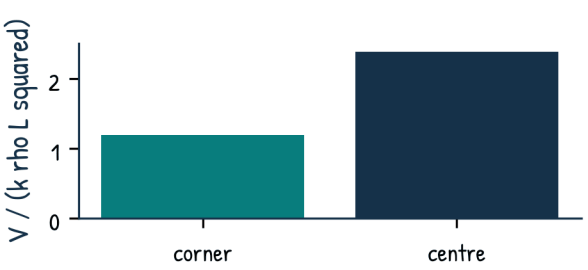
WHY / Check scaling, the centre/corner ratio and mesh refinement together; matching a single decimal is weak validation.

WATCH / The vertex field is not zero: cubic symmetry at the centre is stronger than the limited symmetry at a corner.

Ring centre is a saddle: in-plane V rises while axial V falls



Uniform cube: centre/corner potential ratio two from scaling and tiling



REMEMBER / Analytic limits and convergence tests turn numerical integration into a controlled calculation.

Inverse-square tests and model boundaries

Separate an idealised theoretical signal from a complete experimental inference.

A perturbed exponent needs a length scale

$$F \propto r^{-2}(r_0/r)^{\delta}$$

$$\varphi(r) = \frac{kq}{(1+\delta)r}(r_0/r)^{\delta}$$

The displayed potential differentiates to a force with the specified fixed normalisation. r0 makes logarithms dimensionless. For small delta, expand powers as 1+delta ln(r0/r) while retaining the matching coefficient.

WHY / Changing an exponent without specifying a reference length changes dimensions and hides which coupling constant is kept fixed.

WATCH / The source writes a perturbed potential without the 1/(1+delta) convention factor; its radial first-order shell signal can still be checked, but normalisations must be stated.

Why a shell gives a null test

$$E_{\text{inside}} \simeq -\delta\, kQr/(3R^3), \quad r \ll R$$

For a uniformly charged thin spherical shell and small exponent deviation, direct angular integration gives this leading near-centre field, using coupling normalised at the shell scale. Exact inverse square makes it vanish.

WHY / A cancellation null can be more sensitive to a small law change than direct force comparisons of large signals.

WATCH / A real conducting shell under a changed law can redistribute charge. A complete experimental bound needs apparatus geometry and response, not just the fixed-uniform-shell estimate.

High-voltage dome: useful scaling

$$Q_{\max} \sim E_b R^2/k$$

$$V_{\max} \sim E_b R, \quad U \sim E_b^2 R^3/(2k)$$

An ideal isolated spherical dome reaches the adopted field scale Eb first at its surface. R=1 m and Eb=3 MV/m give Q about 0.334 mC, V about 3 MV and field energy about 501 J.

WHY / The interior of a closed equipotential conductor offers shielding; a belt can transport charge to the dome while its motor supplies work.

WATCH / Real limits depend on supports, curvature, pressure, humidity and leakage. Belt-current factors depend on which charged belt faces and return paths are included.

Atomic and nuclear applicability

$$U_{\text{nucleus}} \simeq \frac{3k(Ze)^2}{5R}$$

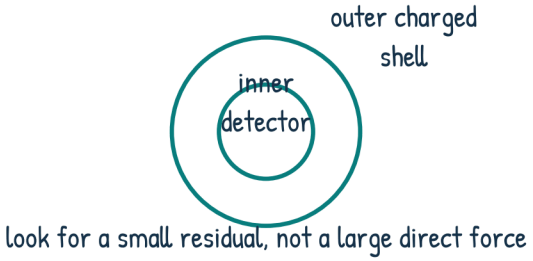
$$K_{\text{contact}} \sim kZ_1Z_2e^2/(R_1+R_2)$$

Uniform-sphere Coulomb energy estimates one repulsive part of nuclear energetics, not the full binding energy. Comparing turning distances with nuclear radii diagnoses when pure Coulomb scattering is no longer adequate.

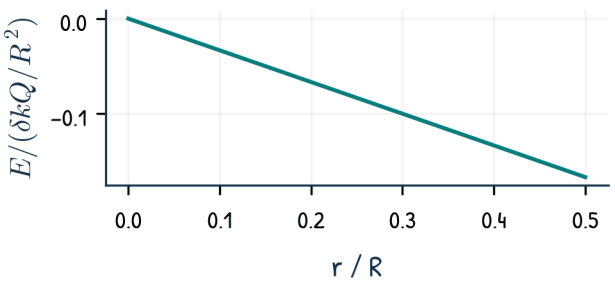
WHY / A successful electrostatic scale estimate can identify where new interactions must enter without modelling those interactions.

WATCH / Quantum charge distribution, nuclear force, screening, recoil and dynamical radiation limit classical point-source extrapolation. No safety or exact material-breakdown claim follows from these estimates.

Schematic shell null test: actual inference needs apparatus modelling



Perturbed inverse square: leading field inside a uniformly charged shell



REMEMBER / Null fields, power-law models and breakdown estimates have assumptions; precision claims need calibration.